

Introduction to gdb

>>We open your eyes to the wonders of the GNU Debugger.

Arise, awake! > by Kshitij Sobti

If you write enough code, sooner or later you'll run into a situation where just putting in print statements to figure out if there are errors won't be enough. You'll need a specialised tool to debug your code. If you're using the free GNU Compiler Collection (GCC), then there's no better tool than 'gdb', also known as the 'GNU Debugger'.

If you use Linux, chances are it's already installed and set up. But it isn't limited only to Linux and Unices. In fact, since it's open source and runs in the command line, it's been ported to all kinds of hardware platforms and operating systems.

If you're wondering why you should use a command line tool in this day and age, well, firstly, that's because it has an extensive feature set, combined with a price point of zero that's hard to match. There are GUI frontends and IDE integrations available for gdb, however, they don't include all of its features. That said, if you just



ddd is a popular GUI frontend for gdb, although at this point the interface is quite dated.

need basic debugging, such tools might work well for you.

Since the GNU Debugger is the most widely available and supported version, it's the command line that we'll look at in this article. So let's begin!

The basic gdb workflow

Before we move on to more complex stuff, let's first get the basic usage of gdb out of the way.

To effectively use gdb, you must compile the source code with debug information included. This isn't strictly necessary, and you can even debug commercial applications for which you don't have the source code.

We can now launch gdb and provide the location to your application. For instance, you can run `gdb myapp` or `gdb /home/user/code/myapp`. This won't immediately run your application in the debugger. Before doing that, you may want to do things such as set up breakpoints.

*New languages



*The programming language by Apple No one's talking about

>>Apple has been known for innovation, but one of its breakthrough that's been under the radar till now is finally getting noticed.

<http://dgit.in/applesPL>



*Google's 'Go' isn't far behind too

>>Google's home brewed language, Go, has some ambitious targets in mind with its project manager aiming to surpass Java in the next 5 years. Whoa! That's some tall order right there, don't you think?

<http://dgit.in/gogogo>



*Facebook's 'Hack' isn't doing bad either

>>Facebook came out with its own programming language based on PHP 'Hack' quite a while ago. And it is making its name in the industry. Hit the link below to know more on its recent developments.

<http://dgit.in/fbhackpl>